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Research paper

The impact of energy use rights trading policy on industrial companies' green investments –Evidence from a quasi-natural experiment in China

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ABSTRACT

As the world's largest energy consumer, China's energy strategy is of great significance in achieving a low-carbon economy for the world. Among the various pathways towards a low-carbon economy, green investment represents a pivotal element. In contrast to traditional investment, which tends to undervalue externalities, green investment considers both economic and environmental effects. This multifunctional approach necessitates the implementation of effective environmental regulation to stimulate green investment. In 2015, China initiated a pilot energy use rights trading policy, offering a quasi-natural experiment in policy for corporate green investment. This paper examines the impact of China's energy use rights trading policy on increasing green investment by Chinese industrial companies. The aim is to provide a valuable reference for the promotion of increased green investment by industrial companies in China and other major economies. The paper empirically examines the micro-economies under the method of multi-period difference-in-differences, taking 4037 observations of 1166 A-share listed industrial companies in China as the research sample from 2013 to 2021, sourced from the Wind database and hand-collected financial statements. It is found that (1) the energy use rights trading policy can significantly promote Chinese industrial companies to increase green investment. (2) After controlling for companies' relevant financial data and undergoing a series of robustness tests, including the placebo test and the propensity score matching difference-in-difference (PSM-DID) test, our main findings remain significant and robust. (3) The heterogeneity results show that the energy use rights trading policy has a more significant role in promoting the increase of green investment in non-state-owned companies, companies in the central region, and companies in high energy-consuming industries.(4) In terms of mechanism analysis, we demonstrate that energy use rights trading policy can promote an increase in green investments by increasing the environmental responsibility index in companies' ESG ratings.

1. Introduction

Since the first industrial revolution, the extensive use of energy has accelerated the economic development of countries all over the world, but it has also brought about serious environmental pollution problems, and countries all over the world have made unremitting efforts to realize harmonious coexistence with nature (Yu et al., 2022, Zhang et al., 2024). As the forerunner of the first industrial revolution, the United Kingdom is fully aware of the threat of energy security and environmental pollution, and put forward the concept of "low-carbon economy" for the first time in the Energy White Paper (2003). The concept of sustainable development encourages countries to reduce their use of coal and oil through technological innovation, scientific and technological development, and the development of new energy sources. This aims to achieve an organic balance between economic development and environmental protection (Liu et al., 2022).

Green investment is a crucial aspect of sustainable development (Siedschlag, 2021; Sheng et al., 2021). It helps to balance economic growth and environmental protection (Musah, 2022) and facilitates the transition to a low-carbon economy (Liu and Zhang, 2022). The United Nations Sustainable Development Goals (SDGs) agenda has emphasized the importance of eco-investments in reducing environmental pollution (Sachs et al., 2019). Countries have called for the resolution of the conflict between economic growth and environmental degradation by focusing on green investments in green industries and technologies (Marshall et al., 2021; Belaïd et al., 2023). As the government and society pay more attention to the low-carbon economy and environmental protection, the focus of companies' strategic decision-making has

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gradually shifted to environmental issues (Chariri et al. 2019).Both the development of environmental protection projects and the improvement of green technologies and equipment require a large amount of green investment, and the increase of enterprises' green investment means that there is a potential headroom for clean production (Sheng et al., 2021), and the ability to increase the share of clean energy in total energy use and thus reduce pollution emissions (Lyeonov et al., 2019) is a key driver for gaining competitiveness (Porter and van der Linde, 1995; Lopez-Gamero et al.; 2010, among others).

Furthermore, achieving a low-carbon economy necessitates the enhancement of theoretical practices in the field of environmental and resource economics (Halkos and Managi, 2023). This discipline prioritizes the creation of clean, comfortable, and aesthetically pleasing living and working environments for humans while minimizing labour consumption through analyzing the conflicts between economic development and environmental protection (Beder, 2011). To achieve this purpose, countries have introduced and developed various environmental regulations (Halkos and Managi, 2016,2017). These include the establishment of energy trading markets, which combines the Coase theorem with environmental and resource economics. The text describes the combination of energy allocation mechanisms with market trading mechanisms, such as "emission rights" markets (Pang and Timilsina, 2021; Xiao et al., 2021; Zhou and Zhao, 2021) and "use rights" markets (Pan and Dong, 2022; Zhang et al., 2021), by utilizing the relationship between transaction costs and property rights arrangements. The emission rights market emphasizes the control of total greenhouse gas emissions, focusing on back-end treatment, such as China's carbon emissions trading (ECRT) (Zhou and Zhao, 2021). While use rights markets focus on limiting total energy consumption and inhibiting energy use behavior on the demand side through front-end governance (Wang et al., 2022; Zhang et al., 2021), such as the white certificates trading system used in Poland, France, the U.S. (Aldrich and Koerner, 2018), and other countries operates differently.

China's energy strategy is of great significance to the realization of the world's low-carbon economy, given its position as the world's second-largest economy, the first industrialized country, and the largest energy consumer. China established its carbon emissions trading market in 2011 and has since achieved significant success in energy conservation and emission reduction (Zhang et al., 2021), elimination of outdated production capacity (Zhu et al., 2017), and promotion of company innovation and employment incentives (X.Yang et al., 2020). The energy use rights trading market was recently established as a unique innovation with Chinese characteristics (Che and Wang, 2022), building on the experience of the carbon emissions trading market (M. Yang et al., 2020; Zhang et al., 2021). However, its effectiveness requires further study (Y.Zhang et al.,2021)). The State Council of China first proposed the establishment of the Energy Use Rights Trading System in 2015 as part of the Overall Program for Reform of Ecological Civilization System. The program also called for the implementation of pilot work for energy use rights trading policy(EURT) in Zhejiang, Fujian, Henan, and Sichuan provinces.

EURT is defined as the behaviour of energy-using entities trading the total energy consumption targets obtained by law. The aim is to limit the total amount of energy consumption in the region (NCRD, 2016). Pilot implementation of EURT began in Zhejiang, Fujian, Henan and Sichuan Provinces in 2015, 2017 and 2018 respectively, as shown in Fig. 1. Since the implementation of the EURT, the growth rate of energy consumption in the four pilot provinces has steadily decreased compared to the years

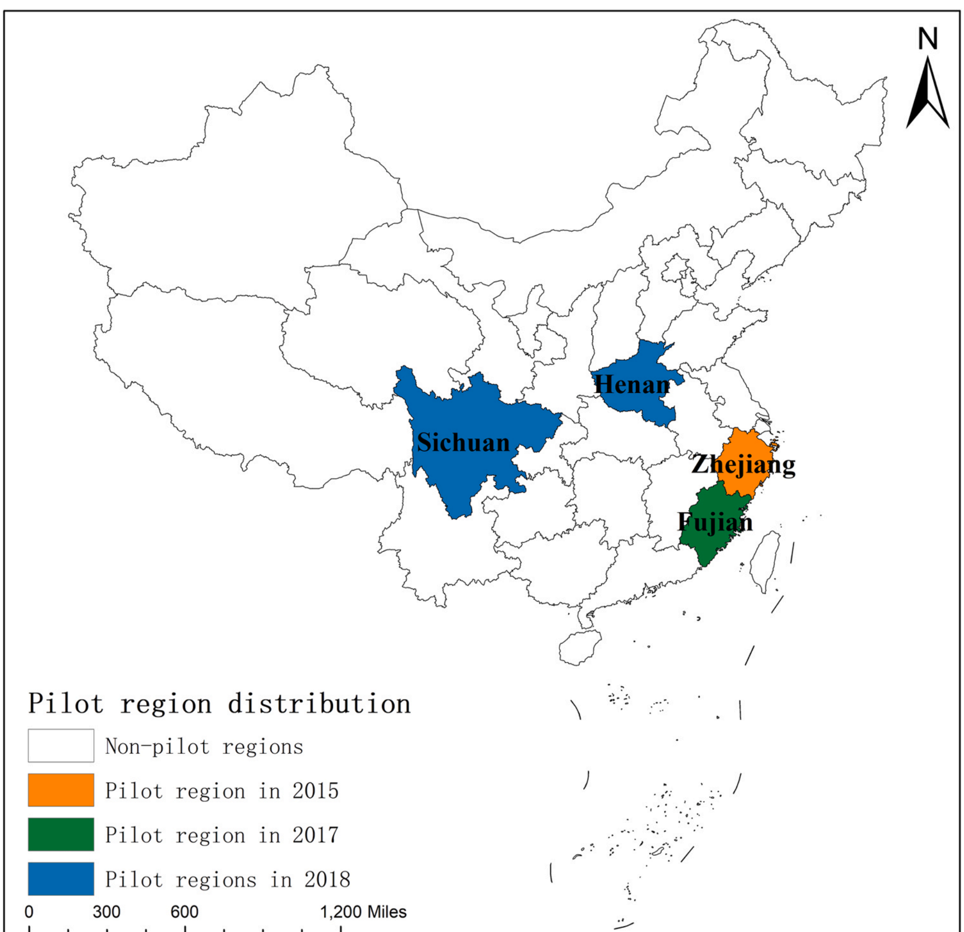

Fig. 1. Pilot provinces of China's energy use rights trading system.

before (Che and Wang, 2022). This indicates the positive environmental impact of the EURT to some extent.

Given the aforementioned context, the objective of this paper is to demonstrate the effect of the EURT on promoting green investments in Chinese industrial companies from a microeconomic perspective, through a multi-period difference-in-differences (DID) approach at the microeconomic level, utilising data from A-share listed companies of Chinese industrial companies for the period 2013–2021. The results can offer timely evidence to inform Chinese policymakers on whether the EURT can be effective in increasing green investments by companies, and whether the impacts of the EURT will change when taking into account the influence of companies' ownership characteristics, regional differences and the nature of the industry. Moreover, the study provides insights into the mechanism of action of the EURT. While this study provides a valuable reference for improvement of Chinese EURT, it also offers transferable insights for other major economies making their low-carbon economy policy choices.

The structure of this paper is outlined as follows: Section 2 provides an overview of the existing literature on companies' green investments and EURT, highlighting the gaps in the existing research. Section 3 outlines the methodology employed to estimate the impact of EURT on corporate green investment. It also details the data capacity, data sources, identification strategy, and potential limitations of the methodology used. Sections 4, 5, 6 and 7 present and analyse the experimental results of the underlying regressions, robustness tests and heterogeneity analyses. They also elucidate the possible impact mechanisms involved. Finally, at the end, findings and insights are summarised to provide new quantitative evidence on the environmental impacts that EURT seeks to achieve.

2. Literature review

2.1. Green investment

Green investment refers to investments that improve environmental quality (Porter and van der Claas, 1995). Under the guidance of the concept of sustainable development, we invest in clean, green, and environmentally friendly production technology and equipment or public facilities to integrate the multiple requirements of the economy, resources, society, and the environment. This allows us to achieve an organic combination of maintaining economic development, safeguarding the health of the population, slowing down the loss of resources, and protecting the ecological environment (Musah et al., 2022; Zhang et al., 2024).

At the macro level, transitioning to a low-carbon economy requires significant investment in green infrastructure, which is not likely to be motivated by the public alone and will require government funding (Meckling et al., 2022; Pekanov and Schratzenstaller, 2023). Currently, countries worldwide are investing in numerous green energy projects. Examples include offshore wind projects in Norway and the Netherlands (Adriaan et al., 2021), onshore and offshore wind power project in Canada (Karayel and Dincer, 2024), and geothermal energy projects in Chile(Vargas-Payera et al., 2020). At the micro level, technological advances play a crucial role in improving resource utilization (Bakhsh et al.,2021). Therefore, an increasing number of companies are investing their capital in research and development projects related to environmental governance, pollution prevention, ecological restoration, green manufacturing, energy saving, and emission reduction (Lopez-Gamero et al., 2010). This approach reduces pollution at the source of company production and has the added benefit of green marketing, resulting in a win-win situation for profitability goals and social responsibility.

Firstly, the motivation of companies' green investments originates from external factors such as financial markets (Managi and Okimoto, 2013; Inchauspe et al., 2015; Ma et al., 2024), economic environment (Saunila et al.,2018) and environmental regulation (Siedschlag and Yan, 2021), as well as internal factors such as the values of company managers (Montalvo,2008), and the nature of companies (Siedschlag and Yan,2021; Haller and Murphy,2012). Regarding external influences, there is a hedging relationship between green and dirty investments in the economic market. According to Managi and Okimoto (2013) and Inchauspe et al. (2015), there is a significant co-movement relationship between net energy stock prices and oil prices. Whenever the price of dirty resources increases, the stock prices of the clean energy market will also rise. This implies that dirty resources can be a suitable option for hedging green investments (Ahmad, 2017). Saunila et al.(2018) empirically analyzed with cross-sectional data from Finnish horse companies and demonstrated the role of economic environment and environmental regulation on companies' willingness to increase their green investments. Considering the heterogeneity of environmental regulations across the world, Siedschlag and Yan (2021) designed a unified and refined economic framework to analyze the effects of internal and external factors on companies' green investments based on empirical tests on micro data from the Irish industrial sector and concluded that environmental regulations, industry competition, and regional spillovers have a significant impact on companies' green investments. In terms of internal influences, Montalvo Corral (2008) argued that the attitudes of business decision makers and societal values would play an important role in investment in clean technologies; Haller and Murphy (2012) concluded that the nature of large, energy-intensive companies has a more pronounced effect on the promotion of companies' green investments, by examining the effect of heterogeneity in company nature on companies' green investments.

Secondly, the existing literature on the impact of companies; green investment focuses on both 'outward' and 'inward' impacts. The 'outward' impact centres around the effect of companies' green investments on the overall climate and national economy. Hung (2023) analyzed the relationship between financial development, green investment, digitization and orderly economic development in Vietnam from 1995 to 2020 using quantile regression, quantile Granger causality, and other methods. The study proved the positive relationship between digitization, green investment and financial development on Vietnam's economic sustainability. Sheng et al. (2021) showed how green investments can coordinate China's economic growth and mitigate environmental pollution using homoskedastic shifted share analysis (HSSA). However, regional and temporal heterogeneity exists, which necessitates the government to establish an eco-compensation mechanism to coordinate green investment growth in different regions. The impact of green investment is analyzed by examining company indicators to draw conclusions about the progress of company development.Since the cost of not disclosing environmental performance is higher than the cost of disclosing green investments (Martin and Moser, 2016), companies tend to opt for green investments to improve their company's financial and sustainability performance measures (Chariri et al., 2019) and create sustainable performance (Saxena and Khandelwal, 2012). Indriastuti et al. (2021) demonstrated the positive impact of green and socially responsible investments on the financial and sustainable performance of manufacturing companies listed on the Indonesian Stock Exchange from 2016 to 2019. The study analyzed financial indicators, green investments, and company social responsibility investments. Li and Li (2022) conducted an empirical analysis of Chinese manufacturing companies and concluded that companies' green investments can enhance the technological innovation capability of companies and improve their level of environmental governance. In summary, the researches suggest that green investments offer companies a means of achieving economic benefits while also contributing to the creation of a more environmentally friendly world.

2.2. Energy use rights trading policy

Since 2015, China has implemented the Energy Use Rights Trading Policy (EURT) and established pilot trading markets in Zhejiang, Fujian, Henan, and Sichuan provinces. However, due to the late start of the

EURT, there is relatively little existing literature on the subject (Pan and Dong, 2022; M.Yang et al., 2020). Previous studies have concentrated on two main areas: evaluating the effectiveness of EURT and examining the relationship between EURT and China's carbon emissions trading (ECRT) in China. On the effectiveness of EURT, previous studies have experimentally verified the positive impacts of EURT in reducing energy vulnerability, energy consumption intensity and structure, low-carbon economic transition, and the impacts of atmospheric pollution intensity (Pan and Dong, 2022; Che and Wang, 2022; Y. Zhang et al., 2021; Yuanguo and Xin, 2023).In addition, Pan and Dong (2023) investigated the coupling effect of fossil energy rights trading (FET) and China's new energy certificate trading policy (RECT) on low-carbon economic development in the context of the policy background of EURT. They assumed that only FET exists in China and, in conjunction with China's renewable energy certificates trading policy (RECT), demonstrated the role of FET and RECT in reducing China's carbon emissions. This study affirmed the importance of promoting the development of renewable energy. It is notable that Y.Zhang et al. (2023) demonstrated that while EURT can provide short-term environmental benefits, it did not achieve the 'Porter effect'. Combining EURT with a carbon tax could mitigate the negative effects of EURT on final output. This suggested that EURT's quota approach and pricing policy would significantly influence the Chinese government's expansion of EURT coverage. Furthermore, EURT shares similarities with ECRT, which was initially proposed in China. Therefore, numerous scholars have analysed and compared the connections and distinctions between the two policies. Wang et al. (2022) compared the difference between individual and joint trading of ECRT and EURT using a Data Included Analysis (DEA) model. The study showed a positive effect of ECRT and EURT on energy saving and emission reduction in a joint trading system. Consequently, there is a need to expand the scope of the EURT pilot and promote the joint application of ECRT and EURT. However, both policies may apply to the same company simultaneously, and dual policy constraints may hinder companies' growth (Wang et al., 2022;Aldrich and Koerner, 2018; Y. Zhang et al., 2022). To address this issue, Y.Zhang et al. (2022) developed a leader-follower game model that combines FCAM to consider coordinating ECRT and EURT under optimal cross-efficiency in Fujian Province. This study suggested strategies for a national coordination mechanism for dual policies.In addition, on top of the above analysis on the impacts of EURT, Pan and Dong (2022) also took Sichuan Province as an example and designed a further improvement model for EURT based on the existing foundation. They used the historical egalitarianism, economic egalitarianism, efficiency allocation model and the game-EFCAM model for energy quota allocation. By simulating with dynamic computational models, they came to the conclusion that the Game-EFCAM model can minimize the resource vulnerability.

It is worth noting that although the EURT was initiated by China, similar 'total energy control' policies exist in other countries, such as the European white certificate system (Zhang et al., 2021). As a tool to overcome the "energy efficiency gap" (Jaffe and Stavins, 1994), white certificates also impose energy saving targets on market participants, which can be offset by companies meeting energy saving targets, and white certificates can be traded (Pavan, 2012, Stede, 2017). Through their application, countries such as the United States, Italy, the Netherlands, and the United Kingdom have made progress in improving employment (Rosenow et al., 2020), closing the industrial energy efficiency gap (Stede, 2017), mitigating global warming (Wirl, 2015), and promoting sustainable energy production (Safarzadeh et al., 2022). Stede (2017), based on a survey of experts conducted during the annual White Certificates Conference in Rome in March 2015, demonstrated that white certificates have an energy-saving and emission-reducing effect, but that they also require stronger government regulation and improved access to finance for companies.Oilonomou et al. (2007) showed that white certificates have been effective in reducing primary energy consumption in Dutch households and improving energy efficiency in the Netherlands, and provide a reference for the introduction of this type of energy-saving policy in the form of total energy limits in countries around the world; Friefrich and Afshari (2016) assessed the usefulness of energy saving obligations and white certificates in the Emirate of Abu Dhabi and found that such investments in energy efficiency can achieve an average return on investment of 6 % in the country over a period of 25 years.They also demonstrated the catalytic effect of the use of white certificates on the development of the energy efficiency market.

However, most of the European countries are developed economies, most of which have peaked carbon emissions by 2020 and are focusing on seeking industrial upgrading and technological innovation (Hultman et al., 2020; Victoria et al., 2020), while developing countries, including China, are still in the stage of economic structural transformation, industrial development and economic expansion, and the formulation of effective environmental policies and sustainable development programmes is critical. (Han et al., 2023).

From the above discussion, the following shortcomings can be summarized: firstly, as a new policy, EURT has achieved some effects and has similarities with the European White Certificate system, which is important for promotion. However, there is still a lack of research on the EURT, and the existing literature on the EURT only takes into account the time when the policy is released at the national level, while there is a difference in the time when the policy is responded to in each province. Secondly, although many scholars have studied the environmental benefits and allocation methods of EURT, there is still a lack of consideration from the level of micro-economies; finally, there is a large body of literature that proves the role of environmental regulation in promoting green investment by companies, but the impact of EURT, an environmental regulation, on green investments by companies has not yet been considered.

3. Methodology and data

3.1. Methodology

The question of the paper is whether the EURT has effectively contributed to the increase of green investments of Chinese industrial companies. The common single difference method can solve the problem by directly comparing the difference in the EURT in the pilot areas before and after the policy. However, this may be inaccurate because certain other factors before and after the EURT can have effects. The single-difference approach cannot account for these differences so all these factors will undoubtedly have an important impact on the EURT then affect the evaluation results. However, the DID method is a natural experiment-based causality evaluation method, which can conduct policy evaluation and mitigate potential endogeneity problems. (Huang and Yang, 2021). A typical DID model comprises two groups: a treatment group and a control group. The policy intervention is the focus of the study, and the policy's effect is embodied by the average treatment effect on the treated (ATT). Therefore, the EURT can be regarded as a quasi-natural experiment.

Since the EURT was initially proposed in 2015, and the pilot provinces gave a gradual response in the following years. Therefore, we employed a multi-period difference-in-differences (DID) model in this paper. The EURT is treated as a quasi-natural experiment, with the assumption that it is exogenous. Given that the pilot policy was implemented in Zhejiang, Fujian, Henan, Sichuan provinces, we match industrial firms from the four provinces to a list of Chinese A-share listed companies. The matched and unmatched companies are designated as 1 and 0, respectively. Consequently, the sample of Chinese industrial companies is divided into two groups: an experimental group (companies in the pilot provinces affected by EURT) and a control group (companies not affected by EURT). At the same time, we take the release of the Interim Measures of Management of Reimbursable Use and Trading of Energy Rights by the pilot provinces as the benchmark. Specifically, 2015 is chosen as the implementation year for EURT in

Zhejiang province, 2017 for Fujian province, and 2018 for Henan and Sichuan provinces.

Therefore, the benchmark model of this paper is formulated as follows:

$$Y_{it} = \alpha_0 + \alpha_1 \mathrm{Treat}_i * \mathrm{Post}_t + \alpha_2 \mathrm{Age} + \alpha_3 Lev + \alpha_4 Roa + \alpha_5 Tagr + \alpha_6 Lr + \mu_i + \lambda_t + \varepsilon_{it} \tag{1}$$

Eq.(1)is a two-way fixed effects multi-period DID estimation model that controls companies fixed effects and time fixed effects simultaneously.

where i and t refer to the company and year, respectively, Y_{it} is the explained variable, 'Treat' represents an individual dummy variable, assuming a value of 1 if the company is situated within the pilot provinces implementing the EURT, and 0 otherwise. The variable 'Post' is a proxy for time. It assumes a value of 0 for the years preceding the formal proposal of the policy by pilot provinces and a value of 1 for the years following the proposal, and the interaction term $Treat_i * Post_t$ was constructed.

We construct a series of control variables, including enterprise age (Age), measured by the length of time the enterprise has been established by 2021; asset-liability ratio (Lev), measured by the ratio of the total debt and total assets of the enterprise; return on total assets (Roa), measured by the ratio of net profit to total assets; total asset growth rate (Tagr), measured by the ratio of current year's total asset growth to the total assets at the beginning of the year; and liquid ratio (LR), measured by the ratio of liquid assets to current liabilities.

The dummy variables μ_i and λ_t represent individual fixed effect and time fixed effect, respectively, while ε_{it} represents the random error term.

3.2. Data

The explained variable is constructed based on the methods of Zhang et al. (2019) and Cao et al. (2023). We manually collate approximately 8000 annual reports from Chinese industrial companies. And the data are then summarized in the note table of construction-in-progress projects, such as flue gas purification, wastewater purification, waste treatment, ecological restoration, dust control, and energy conservation. To enhance the robustness of the regression coefficients in the later paper, the amount of green investments of companies are treated by taking the natural logarithm.The control variables are matched from A-share listed companies of Chinese industrial companies in 2013–2021.

Furthermore, in order to minimize the error, any missing data for variables related to ginger is excluded from the front part of this paper without interpolation.The initial sample was refined by removing companies with questionable financial status, including ST or ST* companies, and excluding samples with missing values. To reduce the influence of extreme values, a 1 % and 99 % trimming treatment is applied to all continuous variables. The data used are obtained from the Wind database, while some companies' data are obtained from the manual compilation of annual reports of listed companies. Given the predominantly cross-sectional nature of the data, this study employs a sample-based approach to adjust the relevant indicators. This process yielded a final sample of 1166 companies and 4037 observations.

In summary, the descriptive statistics of the main variables are shown in Table 1.

4. Empirical results and analysis

4.1. Benchmark regression results

In this section, we examine the concrete impact of the implementation of EURT on the total green investments of the Chinese industrial companies affected by EURT. To this end, we regress model (1) and the results are documented in Table 2. In column (1), we have included individual fixed effects and time fixed effects. The dependent variable is total green investments per company. The results show a significant positive correlation between EURT and total green investments at the 1 % confidence level. In column (2), we introduce the age of the company and observed a statistically significant positive regression result at the 1 % level. In columns (3), (4), (5) and (6), we successively include the asset-liability ratio, the return on total assets, the growth rate of total assets and the current ratio. In particular, the regression coefficient of the multi-period DID consistently shows statistical significance at the 1 % level across all specifications. Furthermore, the results show stability, suggesting that the EURT significantly amplifies the overall green investments of Chinese industrial companies within the designated pilot area.

Table 1
Descriptive statistics.

VARIABLES	(1) N	(2) mean	(3) sd	(4) min	(5) max
y	4037	16.93	2.442	10.73	22.28
Age	4037	18.89	5.277	7	32
Lev	4037	0.459	0.193	0.0698	0.902
Roa	4037	0.0368	0.0553	-0.209	0.183
Tagr	4037	0.182	0.322	-0.270	1.919
LR	4037	1.867	1.722	0.267	10.86
Treat	4037	0.237	0.426	0	1
Post	4037	0.169	0.375	0	1
Treat*Post	4037	0.169	0.375	0	1

4.2. Parallel trend test

The empirical results from the benchmark model show a significant increase in the volume of green investments following the implementation of the EURT. However, the credibility and persuasiveness of this conclusion relies heavily on the validity of the estimation using the multi-period DID method. An essential requirement of the multi-period DID method is that, prior to policy implementation, the trend in green investment should be parallel between the treatment and control groups. This implies that, prior to the implementation of the EURT, there should be no systemic differences in the level of green investment between companies affected by the policy and those not affected by the policy. Therefore, further research is imperative to verify and establish the existence of a parallel trend between the treatment and control groups prior to the implementation of the EURT.

Dummy variables were set for the fourth, third, and second years prior to the implementation of the EURT, and for the first, second, and third years following its implementation. The parallel trend test was conducted to determine if there was a significant difference between the experimental and control groups prior to policy implementation. If the impact coefficient of the explanatory variables fluctuates around 0, it passes the test. The results are presented in Fig. 2. Fig. 2 shows that prior to the implementation of the EURT, the coefficient of the interaction term tended to be negative and significant in the two years leading up to the policy. However, after the policy was implemented, the coefficient of the interaction term increased each year and deviated from 0. This suggests that the EURT has had an impact on the affected enterprises and has gradually created a difference between them and the unaffected enterprises. Our conclusion that the EURT can significantly increase the amount of green investments by companies is reliable.

5. Robustness test

5.1. Robustness test a

To test whether the results of the baseline regression are affected by extraneous policies or unobservable variables, and to strengthen the credibility of our conclusions, The methodology employed in this study

Table 2
The benchmark regression results of multi-period DID method.

VARIABLES	Ln Green investment (1)	(2)	(3)	(4)	(5)	(6)
Treat*Post	0.605***	0.611***	0.610***	0.584***	0.595***	0.590***
	(0.195)	(0.194)	(0.194)	(0.191)	(0.191)	(0.191)
Age		0.134***	0.140***	0.129***	0.126***	0.122**
		(0.0459)	(0.0452)	(0.0490)	(0.0482)	(0.0490)
Lev			1.275***	1.731***	1.581***	1.087**
			(0.437)	(0.451)	(0.450)	(0.514)
Roa				2.367***	1.800**	1.716**
				(0.825)	(0.816)	(0.815)
Tagr					0.336***	0.354***
					(0.107)	(0.108)
Lr						-0.0888**
						(0.0390)
Constant	16.94***	14.39***	13.68***	13.61***	13.70***	14.15***
	(0.0331)	(0.867)	(0.885)	(0.948)	(0.934)	(0.967)
Observations	4037	4037	4037	4037	4037	4037
R-squared	0.682	0.683	0.684	0.685	0.687	0.687

Note: Robust standard errors in parentheses*** $p<0.01$, ** $p<0.05$, * $p<0.1$

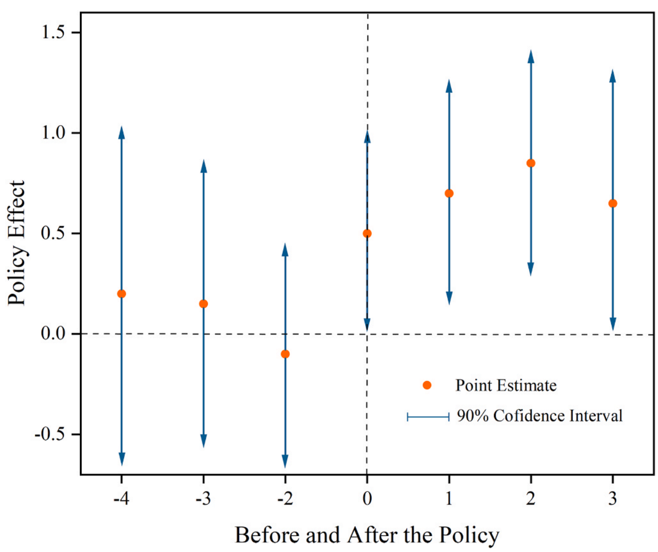

Fig. 2. Results of parallel trend test. Note: The small dots in the figure indicate the regression coefficient β obtained from model (1), and the dotted lines are the 90 % confidence intervals of the estimated values.

is based on the approach proposed by Yu et al. (2021). This entails the artificial setting of time to eliminate intrinsic differences between the experimental and control group companies before and after the pilot implementation of the EURT. It should be noted that the government's policy announcement preceded the actual implementation in the pilot provinces by a significant margin. This circumstance suggested the possibility of anticipatory responses by the pilot provinces prior to policy implementation. We hypothesized that regressions would be run four and three years in advance, respectively. The results showed that the regression coefficient of the interaction term for the dummy policy was statistically insignificant prior to the implementation of the energy use rights trading scheme. The inherent differences between the experimental and control groups had minimal impact on the conclusions drawn in this study and could reasonably be disregarded.

5.2. Robustness test b

In order to test the robustness of the benchmark regression to potential contamination by undisclosed factors, we referred to the method of Yang et al. (2019) and employed a counterfactual methodology. The sample of this paper included 1166 companies, of which 265 were affected by the EURT. Consequently, 265 companies were randomly selected from the 1166 samples and designated as the 'false' treatment group, while the remaining 901 enterprises were regarded as the 'false' control group to construct a dummy variable and then construct an interaction term for the placebo test. Given that the 'false' treatment group was randomly generated, the interaction term of the placebo test should not be significant. That is to say, the coefficient of the interaction term should be zero. In specific, if there is no significant omitted variable bias, the regression coefficients of the false interaction terms should not deviate significantly from zero.

To avoid the interference of small probability events on the estimation results, we used a randomly selected sample and iteratively ran the benchmark regression using formula (1) 500 times, resulting in the derivation of a density distribution plot illustrating the regression coefficients, as shown in Fig. 3. The estimated coefficients exhibit a mean close to zero, which is considerably larger than the baseline regression coefficient (0.590). Additionally, the distribution is more closely aligned with the normal distribution, with the majority of p-values exceeding 0.10, which is not statistically significant at the 10 % level. This indicates that the placebo test is valid, and that the conclusion that the EURT can promote increasing green investments by Chinese industrial companies is not coincidental. This provides evidence of the robustness of the benchmark regression results.

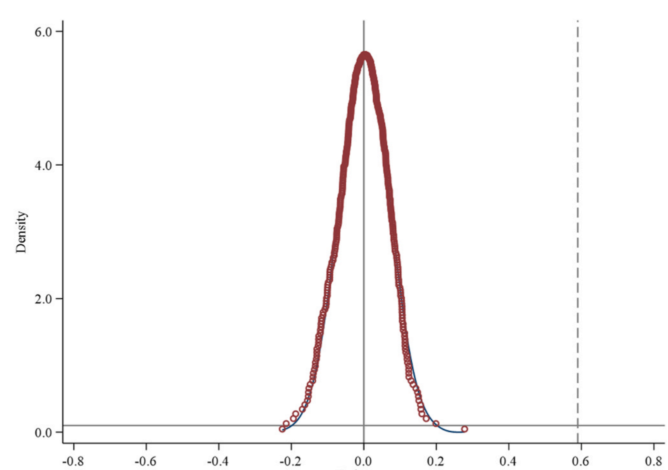

Fig. 3. The result of placebo test b. Note: The vertical line on the right side of the figure represents the true estimated coefficient. The horizontal dashed line at the bottom of the figure represents p = 0.1.

5.3. Further validation based on PSM-DID methodology

It is plausible that there are inherent differences between companies covered by the EURT implementation and those not, raising potential concerns about selection bias. To address the issue, propensity score matching (PSM) was employed to align the experimental group. In order to neutralize the impact of extraneous variables, control variables were incorporated, including enterprise age (Age), asset-liability ratio (Lev), return on total assets (Roa), total asset growth rate (Tagr), and liquid ratio (LR) to verfy the impact of the EURT on green investments. After finding the control group corresponding to the experimental group from 2013 to 2021 through the PSM method and eliminating mismatched samples, the effect of the matched experimental group and the control group was further measured using the DID model.

However, it's important to note that the PSM model is suitable for cross-sectional data, whereas the multi-period DID model is suitable for panel data. To address this challenge, the paper adopts a method proposed by Bai et al. (2022) that specifically targets the incongruence between the suitability of PSM for cross-sectional data and the alignment of multi-period DID with panel data. First, we construct a cross-sectional PSM by treating panel data as cross-sectional data and subsequently matching samples, and second, to address the multi-period DID dilemma in this paper, we use a stepwise matching approach to match samples within each year. Indeed, both methods entail certain limitations. Xie et al. (2021) pointed out that converting panel data into cross-sectional data for processing might lead to the "self-matching" issue, while step-by-step matching could result in inconsistent matching objects before and after the policy implementation. Nevertheless, these two approaches remain valuable research methods given the current circumstances.

Therefore, we set the control variables in this paper, and match the propensity scores using the method of transforming panel data into cross-sectional data and the method of matching period by period. The two data sets are obtained in two ways, and we use them to construct a cross-sectional PSM model, according to the nearest neighbour matching method to find the optimal control group for the companies in the pilot area of the EURT that meet the conditions of the common support, and exclude the part that is not the common support to obtain a new data set. To match the enterprise samples year by year, we used the period-by-period matching method and then vertically merged the matched data of each year to generate the ideal panel data; then, we tested and analyzed the matching effect by testing the two sets of matched data separately by the method of equilibrium test; and finally, we re-estimated the impact effect of the energy rights trading policy on the green investment of companies by applying the multi-period DID method.

Figs. 4 and 5 depict the kernel density plots before and after matching the experimental and control groups, considering panel data as cross-sectional data. From both graphs, it is evident that after matching, the two kernel density curves exhibit minimal deviation, indicating a reasonable efficacy of the cross-sectional PSM method.

Figs. 6 and 7 display the kernel density plots of the experimental and control groups before and after matching using the period-by-period matching method. Comparing these two graphs, it is evident that post-matching, the two kernel density curves almost coincide, indicating that the yearly PSM approach effectively mitigated sample selection bias.

We then ran the regressions separately on the matched cross-sectional data and on the annual data. As shown in Table 3, columns (1) and (2) present the regression results for the multi-period PSM-DID under the two methods, respectively. The results show that regardless of the regression method used, the coefficient 'Treat*Post' associated with 'y' consistently shows a significant positive effect, reflecting no significant deviation from the baseline regression results. To a certain extent, it is marked that the effect of EURT on promoting green investments of companies is robust.

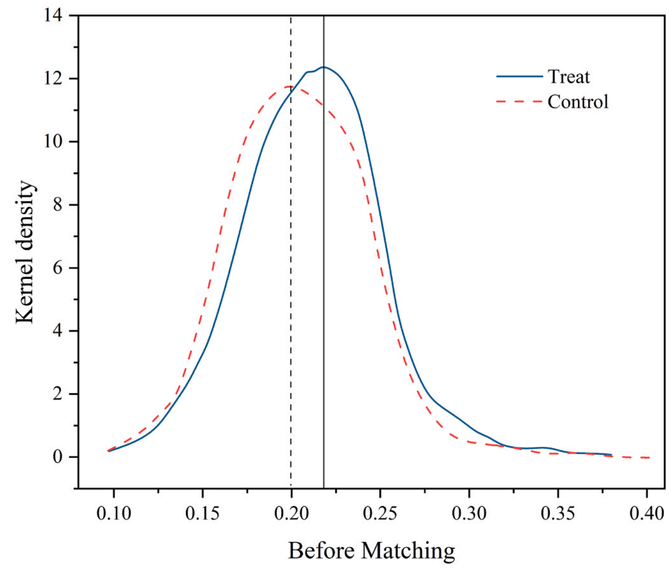

Fig. 4. The kernel density plots before matching the experimental and control groups, considering panel data as cross-sectional data.

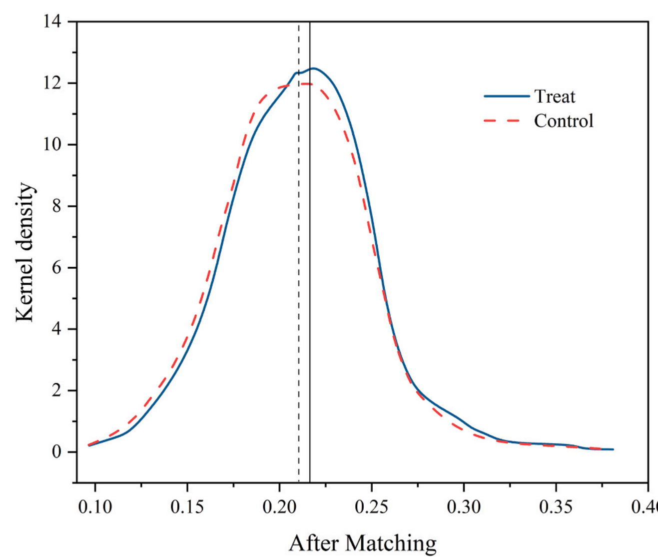

Fig. 5. The kernel density plots after matching the experimental and control groups, considering panel data as cross-sectional data.

5.4. Exclusion of other policy impacts verification

Prior to the implementation of the ERTC, China had already launched a pilot carbon trading scheme (ECRT) in 2011 in six regions, including Beijing, Tianjin, Shanghai, Chongqing, Hubei and Guangdong, and all the regions responded positively and acted positively after the launch of the pilot scheme. As mentioned above, there are many differences between the ECRT and the ERTC, but the implementation of the two policies still partially overlapped in time.Hence, in order to exclude the influence of ECRT on the empirical results, we exclude all industrial companies in the above regions and again conducts regression analysis by multi-period DID method, and the regression coefficients are significantly positive at 1 % confidence level, indicating that after excluding other policy interferences, the EURT has a certain role in promoting companies' green investments.

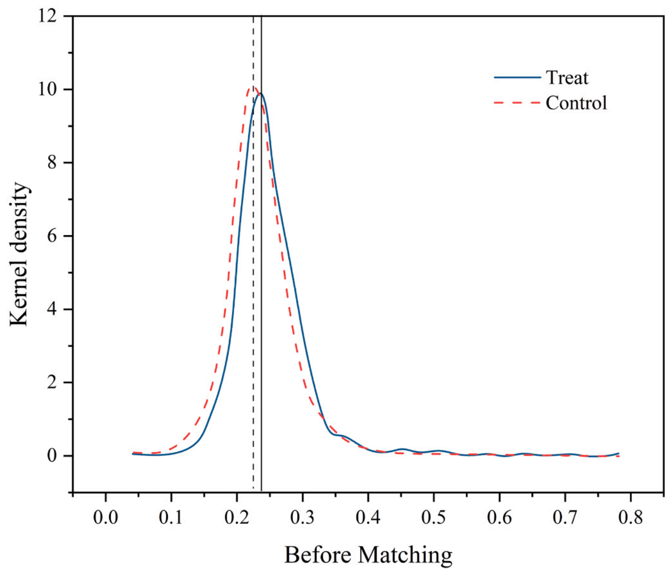

Fig. 6. The kernel density plots of the experimental and control groups before matching using the period-by-period matching method.

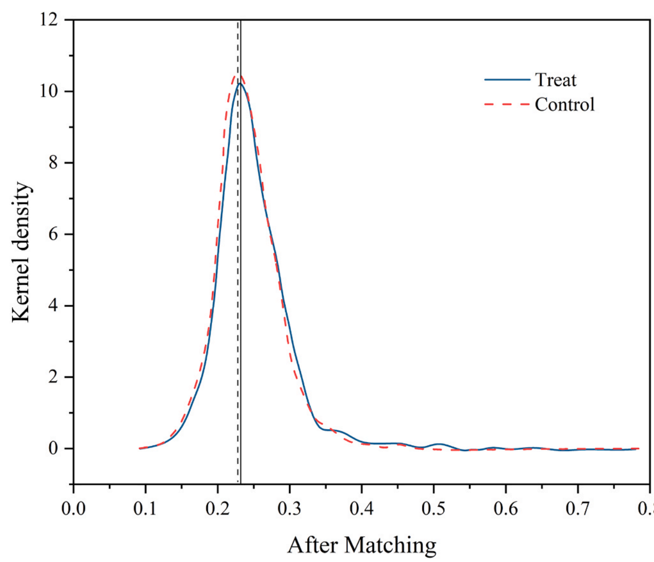

Fig. 7. The kernel density plots of the experimental and control groups after matching using the period-by-period matching method.

Table 3
The results for the multi-period PSM-DID under the cross-sectional data matching and the yearly matching method.

VARIABLES	(1) cross-sectional PSM	(2) Yearly PSM
Treat*Post	0.5634***	0.3885**
	(0.1594)	(0.1633)
Controls	YES	YES
Observations	2087	2079
R-squared	0.6994	0.6847

6. Heterogeneity analysis

This paper's heterogeneity test aims to answer the following questions: Does the promotion effect of EURT on company green investment vary based on the nature of company property rights? Does it exhibit a differentiated effect depending on the city's location? Does it change depending on whether an enterprise belongs to a heavy polluting industry or not? To achieve this, we introduce dummy variables for regression that correspond to the indicators of companies in the four pilot provinces. For state-owned companies in the pilot provinces, the value is 1, and 0 for non-state-owned companies. The value is 1 for companies in the eastern region, 2 for those in the central region, and 3 for those in the western region. For companies in the heavy pollution industry in the pilot provinces, the value is 1, and 0 for those in the non-heavy pollution industry.

Table 4
The results of the mechanism analysis.

VARIABLES	(1) y	(2) E-Score	(3) y
E-Score			0.0153***
			(0.00259)
Treat*Post	0.468***	1.976***	0.437***
Constant	16.94***	14.39***	13.68***
	(0.0331)	(0.867)	(0.885)
Company FE	Y	Y	Y
Year FE	Y	Y	Y
Controls	Y	Y	Y
Observations	4037	4037	4037
R-squared	0.682	0.683	0.684

6.1. Heterogeneity in the ownership characteristics

The differing levels of responsiveness of enterprises of different ownership to investment activities have been a topic of significant interest and debate among domestic researchers. However, it remains unclear whether state-owned enterprises are more inclined to increase their green investment activities or whether non-state-owned enterprises are more motivated to do so.

From columns (1) and (2) in Table 5, it can be seen that the effect of the EURT on promoting green investments is significant for non-state-owned enterprises at the 1 % level and for state-owned enterprises at the 10 % level. Furthermore, it can be concluded that non-state-owned enterprises are more active and positive towards green investment than state-owned enterprises. We posit that the primary reason for this discrepancy lies in the fact that non-state-owned enterprises, due to indirect state control and government involvement, possess a greater degree of autonomy in management and operation than state-owned enterprises. This autonomy allows non-state-owned enterprises to exercise greater choice in investment and financing options.

6.2. Heterogeneity across regions

It is worth noting that the EURT was launched in China's four provinces of Zhejiang, Fujian, Henan, Sichuan pilot work, these four provinces are located in China's eastern, central and western regions, provinces and different locations means that the level of economic development, regional resources, energy conservation and emission reduction basis of the differences, which will also make the cost of green behaviour of companies in different regions will vary, so the degree of energy trading policy for the degree of green investment in different areas of the enterprise will be different.

Table 5 shows that the central region (Henan) exhibits significantly higher motivation to increase green investments compared to the western region (Sichuan) and the eastern region (Fujian, Zhejiang). There are two possible reasons for this difference: firstly, the length of the policy impact. For regions like Zhejiang and Fujian, which have had a longer period of time for the EURT or already have a certain level of comprehensive energy utilization before the policy is implemented, their potential for improving energy allocation is relatively limited. This also suggests that their endogenous elasticity will be less than that of companies within regions with a shorter length of time for the policy

Table 5
The results of heterogeneity analysis.

VARIABLES	(1) SOEs	(2) Non-SOEs	(1) East	(2) Central	(3) West	(1) High	(2) Low
Treat*Post	0.575*	0.733***	0.418*	1.188**	0.891*	0.826**	0.533**
	(0.313)	(0.239)	(0.228)	(0.521)	(0.525)	(0.338)	(0.231)
Constant	14.69***	16.14***	15.18***	18.97***	16.52***	14.91***	16.07***
	(1.453)	(0.883)	(0.812)	(2.766)	(2.138)	(1.301)	(0.933)
Company FE	Y	Y	Y	Y	Y	Y	Y
Year FE	Y	Y	Y	Y	Y	Y	Y
Controls	Y	Y	Y	Y	Y	Y	Y
Observations	341	694	688	173	174	347	661
R-squared	0.153	0.056	0.076	0.160	0.058	0.124	0.056

Note: Standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

pilot. Secondly, the economic context. China's regional development is characterised by significant imbalances. These are a consequence of differences in resource endowments and policy support, which have resulted in substantial disparities in development between regions. Overall, the eastern region of China is the most developed, followed by the central region, and the western region is relatively slow to develop. The favourable economic conditions also indicate that the region has experienced rapid economic expansion, which has resulted in the development of more advanced infrastructure. This implies that there is less room for improvement.

6.3. Heterogeneity in the nature of the industry

The EURT aims to regulate the total energy consumption in the region by controlling the energy amount consumed by companies. The Pilot Programme for the Compensated Use and Trading System for Energy Use Rights stipulates that each pilot region should make a clear distinction between high-energy-consuming industries and low-energy-consuming industries and implement classified guidelines. Therefore, the EURT has varying impacts on companies based on their energy consumption levels.

Table 5 shows that companies with high energy consumption have a greater interest in increasing their green investments. High energy-consuming industries are often characterized by high energy consumption and emissions, making them a challenge for China's carbon neutrality target. As a result, these industries face stricter government regulations and higher production costs, which creates an urgent need for them to improve their green productivity and promote green transformation. In contrast, low energy-consuming companies have surplus energy consumption and a lower demand for green investments.

7. Mechanism analysis

The above empirical results provide a set of robust and consistent evidence that EURT can promote increased green investments by industrial companies, but how to find the factors that guide and encourage companies to increase green investments is a question that needs to be answered urgently. Company environmental responsibility may play a crucial role in increasing green investments and improving environmental governance by companies.

In recent years, under the guidance of a series of policies with the keyword "green", green has gradually become the background colour for the development of industrial companies in China, which means that companies should not only focus on their profitability, but also gradually build the awareness of environmental responsibility, improve the environmental governance from the inside of the enterprise, and then increase the intensity of green investment. However, it is not clear whether company environmental responsibility affects the relationship between EURT and company green investment. So we further explore the role of company environmental responsibility (EL) index in the EURT and companies' green investments. For this purpose, we introduced the E-score from the company's ESG score as a mediator and tested it using stepwise regression.

We matched the data at the company level. The fundamental data on companies was sourced from the Chinese rating agency Huazheng, which is a leading institution in the field of Chinese ESG studies (Wang and Liu, 2024). And then we introduced the E-score from the company's ESG score as a mediator, with samples containing missing values excluded. After data matching,we accumulated 4211 observations covering 1110 Chinese industrial companies from 2013 to 2021. And other indicators are consistent with the above.

The validation steps are as follows: The EURT dummy variable is regressed with the E-score, and if the regression coefficient is significant, it means that EURT has an effect on the increase of company environmental responsibility; then the E-score is regressed with the company green investment data in the next step, and if the result is significant, it means that company environmental responsibility has an effect on company green investment; finally, the EURT dummy variable, the E-score and the company green investment data are entered into the regression model together, and if the coefficient is decreasing or insignificant, it means that such an effect is partly or wholly due to the implementation of the company environmental responsibility index.

Following the above steps, we built the following model:

The first step is to verify the impact of the EURT on the green investment of companies.

$$Y_{it} = \alpha_0 + \alpha_1 Treat_i * Post_t + \alpha_2 Age + \alpha_3 Lev + \alpha_4 Roa + \alpha_5 Tagr + \alpha_6 Lr + \mu_i + \lambda_t + \varepsilon_{it} \quad (2)$$

The second step is the verification of the impact of the company environmental responsibility index on companies' green investment.

$$E - Score = \beta_0 + \beta_1 Treat_i * Post_t + \beta_2 Age + \beta_3 Lev + \beta_4 Roa + \beta_5 Tagr + \beta_6 Lr + \mu_i + \lambda_t + \varepsilon_{it} \quad (3)$$

In the third step, the model included the EURT dummy variable, the company environmental responsibility index, and the company green investment simultaneously.

$$Y_{it} = \gamma_0 + \gamma_1 E - Score + \gamma_2 Treat_i * Post_t + \beta_2 Age + \gamma_3 Lev + \gamma_4 Roa + \gamma_5 Tagr + \gamma_6 Lr + \mu_i + \lambda_t + \varepsilon_{it} \quad (4)$$

The Table 4 shows that the coefficient of the interaction term in column (1) is significantly positive at the 1 % level, indicating that EURT does positively affect green investments in industrial companies. Column (2) shows that the regression coefficient of the interaction term is significant at the 1 % level. Furthermore, the coefficient of E-score is significant at the 1 % level, as shown in column (3). Also,the coefficient of the interaction term decreased when the EURT dummy variable, the company environmental responsibility index, and the companies' green investments were included in the model together, implying that company environmental responsibility plays a mediating role between the EURT and companies' green investments.

It can be assumed that the EURT can improve company environmental governance by increasing company environmental responsibility awareness, which in turn encourages companies to increase green investments.

8. Discussion

Using panel data of A-share listed corporations from 2013 to 2021, the paper examines whether EURT can encourage green investments in industrial companies and the moderating effect of company environmental responsibility in this relationship. This study provides strong support for the hypothesis through empirical analyses using the multi-period DID method. As follows, this study discusses the main results.

First, the EURT has significantly increased companies' green investments. For a wide range of other major economies, it is crucial to develop effective environmental regulations to ensure that industrial expansion and environmental protection go in harmony. This study confirms a practical solution, which is to encourage industrial companies to consciously follow a green development path. The study results indicate that industrial companies invest their capital in research and development of environmental management, ecological reconstruction, and low-carbon manufacturing programs with the intensive implementation of the EURT. There are two possible reasons for this: firstly, the EURT creates constraints on the total amount of energy used by industrial companies. Excessive energy usage can result in higher purchasing costs, which may compel industrial companies to make green investments, such as energy conservation at the production source. Additionally, the EURT is integrated with market mechanisms, which can lead to excess returns for industrial companies that conserve energy. Therefore, the EURT can motivate companies to increase green investments.

Second, the heterogeneity analysis shows that there are differences in the impact of the EURT on companies' green investments. Regarding the type of companies, non-state-owned companies have a stronger incentive to increase green investment than state-owned companies. This finding proves that business freedom is one of the driving forces for increasing green investments, i.e. the lower the intensity of government regulation, the greater the motivation to increase green investments. Therefore, we should be aware that increasing green investments requires industrial companies to gain more autonomy. In terms of regional characteristics, the pilot effect of the EURT is most evident in the central region, then in the western region and finally in the eastern region. This result shows a positive correlation between the level of economic development and green investments under the EURT. In addition, the policy effect is more significant in regions with more room for improvement in energy facilities, which also proves the positive effect of the EURT in promoting the green transformation of industries from another perspective. Regarding the type of industry, EURT has a more significant effect on increasing green investments in high-energy-consuming industries and is relatively weaker in low-energy-consuming industries. This finding suggests that the incentive effect of the market mechanism is weaker than the constraining effect of the penalty mechanism in terms of policy effects.

Third, company environmental responsibility plays a mediating role between EURT and green investments by industrial companies. This finding explains how EURT promotes increased green investments by industrial companies from the perspective of mechanism analysis. In recent years, the growing concept of environmental protection has gradually changed the attitudes and behaviour of company managers. The implementation of EURT has had a positive effect on the establishment of environmental responsibility of industrial companies, which in turn has contributed to the increase of their green investments.

The theoretical implications of this study include two aspects. On the first, previous studies have examined the effectiveness of EURT and the relationship between EURT and other environmental regulations at the macro level, but few have examined the role of EURT on company behaviour at the micro level. To fill this gap, this study constructs a micro-company based research framework to explore the relationship between EURT and companies' green investments, complementing the existing literature. This innovative framework allows us to understand the impact of environmental regulation on companies' behaviour at the micro level, thus contributing to a deeper understanding of the policy and searching for its shortcomings.On the second, this study further deepens the understanding of EURT from a mechanism analysis perspective. EURT not only directly influences industrial companies to increase their environmental investments, but also exerts an indirect effect through the increase in company environmental responsibility. Previous studies in the literature have not explored the relationship between EURT and company ESG indicators. The data for company environmental responsibility in this study are derived from the 'E' indicator in ESG indicators. Linking EURT to the 'E' indicator in ESG indicators not only addresses the shortcomings of previous studies but also enhances the understanding of the effectiveness of EURT. Additionally, it explains the channels through which EURT influences the increase of green investment by industrial companies.

The policy implications of this study include three aspects as follows. Firstly, the results of the study confirm the positive effect of EURT, i.e. EURT can effectively increase the number of green investments by industrial companies. Therefore, it is necessary to summarize the pilot experience in a timely manner before gradually expanding the scope of the pilot to other regions in China. In detail, the government can choose to prioritize provinces with weak energy saving and emission reduction bases, and supplement companies in the pilot provinces with PPP government subsidies, tax incentives, loose monetary policy and other macro-control tools. Secondly, the results of the heterogeneity analysis in this study show a clear differentiation feature, which means that there is a need to refine the preferential policy in different regions and the differentiated regulations in different industries. Therefore, more refined regulations and support policies should be proposed for regions with different economic bases and different types of industries which will improve the effectiveness of policy implementation and have a positive impact on stimulating the enthusiasm of companies of different types and regions.Thirdly, the results of the study show that company environmental responsibility is the key channel for EURT to positively influence the green investments of industrial companies. Therefore, policy makers should formulate related policies (e. g., including the setting of environmental standards in the conditions for granting energy use rights, etc.) to continuously improve the initiative of environmental responsibility of industrial companies whereas adopt incentive measures (e. g., including the environmental performance of companies as one of the factors in the pricing of energy use rights and giving certain rewards to companies with good environmental responsibility performance, etc.) to reduce the cost of companies to bear environmental responsibility.

In conclusion, the marginal contribution of the study lies in following aspects. First, we demonstrate the effect of energy trading policy to promote green investments of Chinese industrial companies from the perspective of microeconomy. Secondly, we added a literature chain on the heterogeneity of green investments effects of companies of different nature under the energy right trading policy, providing a more accurate and differentiated reference for further improving the EURT, and also providing a theoretical basis for the expansion and construction of the energy right trading market. Finally, we further reveal the possible influencing mechanism, providing new quantitative evidence for the environmental effects of energy trading policies. In addition to the research on the impact of EURT, it also enriches the antecedent research on the promoting effect of environmental regulations on green investments.

Despite the contributions of this study, it is not without limitations,. that require further exploration in future research. Firstly, the research sample of this study includes only A-share listed companies due to the availability of panel data. However, the non-listed firms are excluded from the data set, which may lead to sample selection bias. In the future, further reaearch can expand the sample size or use a sample of unlisted firms. Secondly, EURT's effect on promoting company environmental responsibility, which in turn increases companies' green investment, is not limited to the mechanism by which the EURT affects companies' green investment. Thirdly, while the EURT may increase green investment, there is still a lack of research on the level of green investment and its subsequent effects. It is important to determine whether there is a significant change in the environmental performance or productivity of companies after they increase their level of green investment, as well as the overall effect on society. Therefore, future research should further explore the benefits of micro-economies and the realization of social benefits. Furthermore, the research in this paper informs the improvement of China's energy use rights policy, while China's experience provides a pathway for other major economies that are considering upgrading their corporate green investments. However, a focus on the Chinese context may limit the universality of the findings to other economies with different policy frameworks. Consequently, further research is required to ascertain the generalizability of the findings.

CRediT authorship contribution statement

Jiaxin Li: Writing – original draft, Software, Methodology, Conceptualization. **Yiwei Guo:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare no conflict of interest for the order and cooperation.

Data Availability

Data will be made available on request.

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